

System Blueprint

TNPG: financebros

Project: Blossomberg Terminal

Target ship date: 2026-06-01

Roster

Name	Email	Primary Role	Secondary Role
Haowen Xiao	haowenx2@nycstudents.net	Frontend, UI, charts, deployment, page-layout	Project Manager, team coordination, deadlines
Cody Wong	codyw2789@nycstudents.net	Database, API integration	Stock datasets, backend
Robert Chen	robertc295@nycstudents.net	Backend, Flask app, routing, request handling	Authentication, portfolio pages, user accounts
James Sun	james9845@nycstudents.net	Machine learning forecast, AI analysis	LLM integration, Review

Summary:

Blossomberg terminal is a simplified-down and modernized clone of the Bloomberg Terminal that uses live stock data, charts, and ML forecasting with an AI analysis layer for market exploration and learning

Problem being solved:

The real Bloomberg Terminal costs almost \$25,000 / year and has a UI that was built in 1980s that is very intimidating for beginners. Retail traders and students who want to learn can use this to learn more about stocks and other features of the stock market before deciding if they need a full Bloomberg terminal.

Target Users:

- Stock traders
- Aspiring Students learning finance
- Retail / hobby investors (“finance bros”)

Why this project matters:

This project matters because Bloomberg can be both very overwhelming and expensive to a normal user or someone who wants to get into this type of learning. We hope that this project can help students learn but also for professionals to use for their day to day lives.

MINIMUM VIABLE PRODUCT (MVP) SCOPE:

Core features (Required for Final Submission):

- Stock data pulling live and historical prices from Yahoo Finance API or cached in SQLite
- Dashboard with searchable stock tickers, growth charts, comparison graphs, and volume charts using [Chart.js](#)
- ML forecast page using Scikit-learn regression models trained on the FAANG historical dataset.
- Portfolio / watchlist system letting users save tickers to their account and view their performance
- AI Analysis panel using OpenRouter that answers questions like “Why did NVDA rise today?”
- Stock viewer page (/stockviewer): table of top 15 stocks showing live price, change %, open, high, low, volume, and 30-day charts
- Stock detail page: Gives all important data about a specific stock such as their valuation, financials, dividends, analyst ratings, etc with a extended graph and Sharpe ratio comparison.
- Simulated portfolio with paper trading: start with \$100,000 cash and buy/sell semi-live yfinance prices with transaction history and portfolio reset

Stretch Features (Only if MVP is Complete)

- Candlestick and line graphs
- Sector and index heatmap
- Email/alert system for prices

Explicit Non-goals

- Personally learn more about the FinTech industry
- Learn more about stocks and equations used

Features intentionally excluded:

- Real-money trading or integration of any kind
- Tick-by-tick streaming data
- Options, futures, crypto, forex

Technology stack:

Layer	Selected tool
Backend framework	Flask
Frontend framework	TailwindCSS, JavaScript
Database	SQLite
Authentication	Flask sessions
Orm/DB library	N/A
Data Visualization	Chart.js
Machine Learning	Sci-kit-Learn
Market Data	yFinance
AI	OpenRouter API

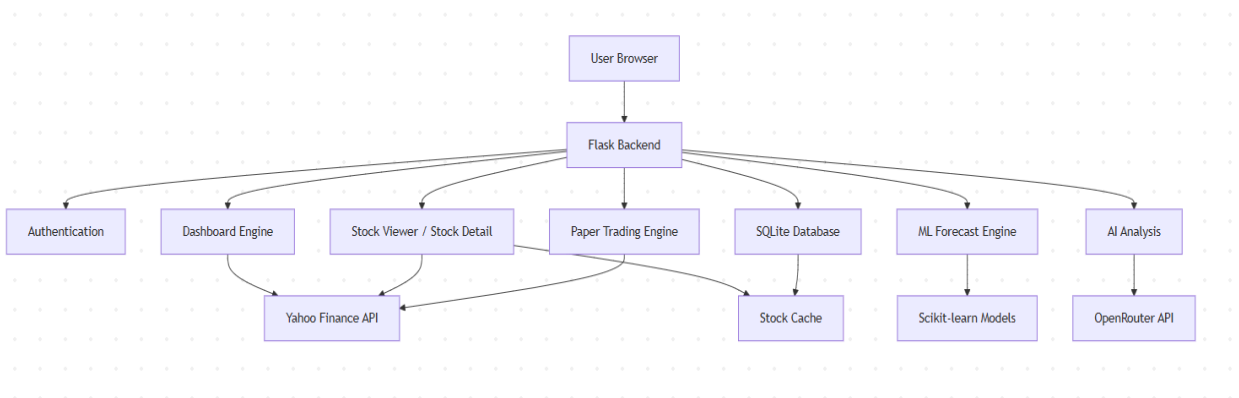
Why this stack was chosen:

Flask is what everyone has been using since day 1; it works well and fits our needs since we're mostly serving cached data and proxying API calls instead of handling a lot of backend logic. SQLite is because our dataset fits in a single file and we don't expect many users writing to the database at the time of MVP. We decided to go with TailwindCSS because it is the most customizable and easiest to work with for building a dashboard, and Chart.js is the best option for our needs for graphs, comparisons, and volume visualizations for equities as if we finish the MVP. OpenRouter was chosen for the AI layer because it gives us access to multiple LLM providers without separate billing, and the student pack credits make it free for this project.

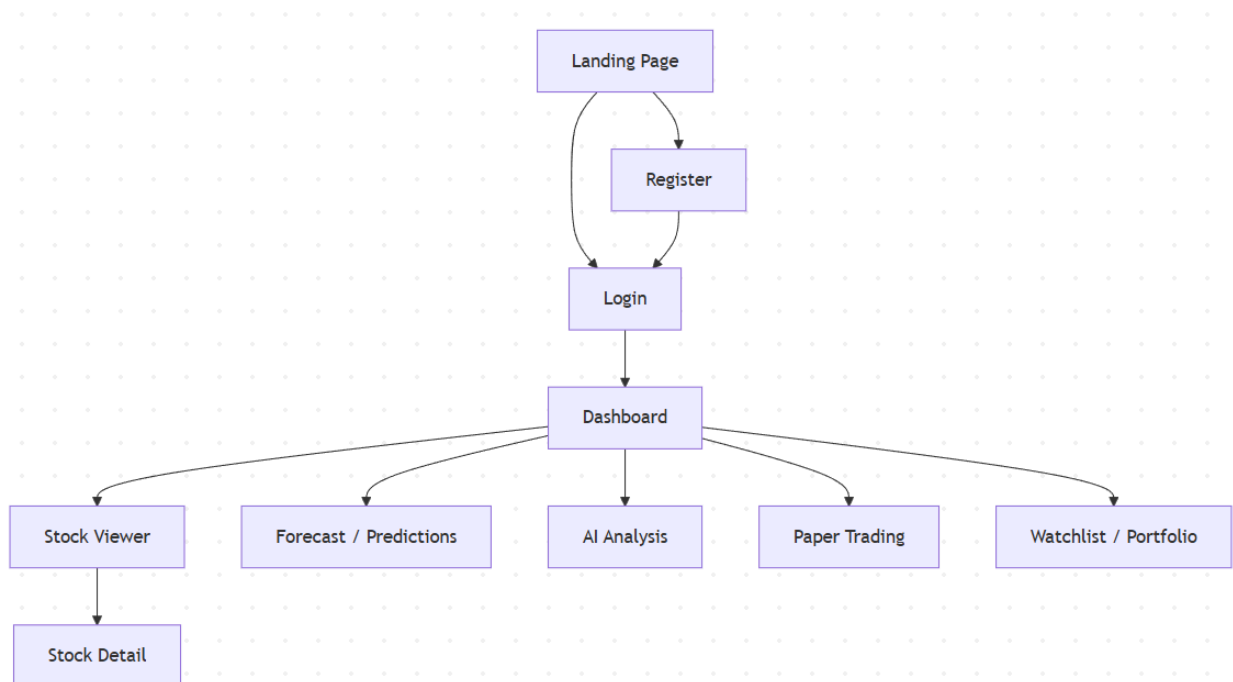
Team Ownership Plan

Team member	Primary Ownership	Secondary Ownership	Specific Deliverables
Haowen Xiao	Frontend, UI Infrastructure	Project Management, Deployment	Build dashboard UI, implement navigation/layout, integrate charts, and manage deployment setup
Cody Wong	SQLite, API integration	Stock datasets, backend	Set up SQLite database, integrate stock APIs, cache historical data, and manage backend data flow
Robert Chen	Flask, routing, sessions	Authentication, Portfolio pages	Build Flask routes, manage user sessions/login system, and develop portfolio/watchlist pages
James Sun	ML forecasting system, AI integration	Flask backend support, analytics features	Train and evaluate Sk-learn models using stock data from yFinance, packaging model weights and integrating into Flask app to create prediction-vs-actual comparison charts.

Component map



Site map



Key User Stories

Day-trading retail investor

As a retail investor actively trading during market hours, I want to pull up a ticker, see its growth and make a decision. I want real-time price updates, technical indicators, and AI-generated summaries with verified news sources so that I can react quickly to market movements and make fast trading decisions.

Finance student

As a finance student learning how markets work, I want to compare equities side by side and view prediction models with confidence intervals and historical trend explanations so that I can better understand how market behavior and forecasting models operate before investing real money.

Hobbyist with a watchlist

As a casual investor managing a personal watchlist, I want to save and organize my favorite tickers and monitor their long-term performance in one dashboard so that I can conveniently track my investments without using multiple platforms.

Database Design

Table Name	Purpose
users	Stores user account information for login and sessions
watchlist	Stores the stock tickers each user saves
stock_cache	Stores cached stock price data pulled from yFinance
portfolio	Stores simulated user holdings for tracked stocks
ai_queries	Stores recent AI analysis questions and answers
ml_predictions	Stores ML forecast results for selected stocks

USERS		
Type	Field	Description
INTEGER PRIMARY KEY	id	Unique user ID
TEXT UNIQUE	username	User login name
TEXT UNIQUE	password	User password

WATCHLIST		
Type	Field	Description
INTEGER PRIMARY KEY	ID	Unique watchlist item id
INTEGER	User_id	Links to user id
TEXT	Ticker	Saved stock ticker

stock_cache		
Type	Field	Description
INTEGER PRIMARY KEY	id	Unique stock data row
TEXT	ticker	Stock ticker symbol
DATE	date	Trading date
REAL	open	Opening price
REAL	high	Highest price
REAL	low	Lowest price
REAL	close	Closing price
INTEGER	volume	Trading volume

Portfolio		
TYPE	FIELD	DESCRIPTION
INTEGER PRIMARY KEY	id	Unique portfolio entry
INTEGER	user_id	Links to users.id
TEXT	ticker	Stock ticker owned
REAL	shares	Number of shares
REAL	avg_price	Average purchase price
DATETIME	created_at	Time holding was added

AI_QUERIES		
TYPE	FIELD	DESCRIPTION
INTEGER PRIMARY KEY	id	Unique AI query ID
INTEGER	user_id	Links to user.id
TEXT	ticker	Related stock ticker
TEXT	question	User's question
TEXT	answer	ai-generated response
DATETIME	created_at	Time question was asked

Ml_predictions		
TYPE	FIELD	DESCRIPTION
INTEGER PRIMARY KEY	ID	Unique prediction ID
TEXT	ticker	Stock ticker
DATE	prediction_date	Date prediction was made

Ml_predictions		
DATE	target_date	Date was being predicted
REAL	predicted_price	ML-predicted stock price

paper_balance		
TYPE	FIELD	DESCRIPTION
INTEGER PRIMARY KEY	user_id	Links to user.id
REAL	cash	Current cash balance (default \$100,000)

paper_holdings		
TYPE	FIELD	DESCRIPTION
INTEGER PRIMARY KEY	user_id	Links to user.id
TEXT	ticker	Stock ticker
REAL	shares	Shares currently held
REAL	avg_cost	Average purchase price

paper_balance		
TYPE	FIELD	DESCRIPTION
INTEGER PRIMARY KEY	user_id	Links to user.id
TEXT	ticker	Stock ticker
TEXT	action	BUY or SELL

paper_balance		
REAL	shares	Shares traded
REAL	price	Price per share at action
REAL	total	Total value
DATETIME	created_at	Timestamp of trade (UTC)

Testing Plan

Component	What to Test	Testing Method	Success Criteria
Stock Data API	Ticker search, live prices, historical prices	Test with common tickers like AAPL, NVDA, TSLA, and invalid tickers	Correct data loads without crashing, invalid tickers show an error message
SQLite Database	Users, watchlists, cached stock data, predictions	Add, update, delete, and retrieve sample records	Data saves correctly and can be retrieved after refreshing
Flask Routes	Page routing and backend API endpoints	Manually visit each route and test API responses	All routes load correctly and return expected data
Authentication	Login, logout, user sessions	Create test accounts and try valid/invalid logins	Users stay logged in correctly and cannot access private pages when logged out
Dashboard UI	Layout, stock search, charts, responsiveness	Test on desktop and smaller browser widths	UI is readable, organized, and does not break on resize
Chart.js Visualizations	Price charts, comparison graphs, volume charts	Compare displayed chart data with backend data	Charts match the correct ticker data and update when users search
Watchlist / Portfolio	Save tickers, remove tickers, view performance	Add and remove sample tickers from a test account	Saved tickers stay attached to the correct user

ML Forecasting	Prediction output, prediction-vs-actual chart	Run model on historical yFinance data and compare output format	Forecast loads, graph displays correctly, and model does not crash
AI Analysis Panel	User questions and AI stock explanations	Ask sample questions like “Why did NVDA rise today?”	AI returns relevant explanations and handles missing data gracefully
Error Handling	Invalid inputs, API failures, empty results	Test fake tickers, bad login info, and disconnected API cases	App shows clear error messages instead of crashing
Final Integration	Full user flow across the app	Login, search stock, view chart, save watchlist, run forecast, ask AI question	Full workflow works smoothly from start to finish

Timeline

Week 1:

- Create GitHub repository and organize overall project structure
- Develop login/logout system and authentication flow
- Set up Flask backend framework and routing system
- Create SQLite database tables and schema
- Integrate yFinance APIs for live for historical stock data
- Design starter dashboard UI using TailwindCSS
- Set up Flask sessions and backend infrastructure
- Ensure the application runs locally with working routes and stock data retrieval

Week 2:

- Build dashboard interface and responsive layouts
- Implement stock search functionality
- Create Chart.js price, comparison, and volume charts
- Develop login/logout system and authentication flow
- Build watchlist and portfolio pages
- Implement stock data caching system
- Train and test Scikit-learn forecasting models
- Begin connecting ML prediction outputs to the Flask backend
- Ensure users can search stocks, view charts, and save watchlists

Week 3:

- Integrate ML forecasts into frontend prediction charts
- Research and preprocess FAANG historical datasets for ML forecasting

- Build AI analysis panel using OpenRouter/OpenAI APIs
- Improve dashboard styling and responsiveness
- Perform backend integration and API reliability testing
- Fix bugs and optimize application performance
- Conduct full end-to-end testing across all components
- Prepare deployment and production setup
- Finalize presentation, documentation, and demo materials
- Complete final polishing before the June 1 submission deadline

Internal Deadlines:

May 15: Complete Flask app structure, GitHub repository setup, and overall project organization.

May 17: Finish SQLite database tables and backend data structure.

May 18: Complete yFinance API integration and stock data caching system.

May 20: Finish dashboard UI, stock search, and Chart.js visualizations.

May 22: Complete authentication, sessions, and portfolio/watchlist pages.

May 25: Train and integrate ML forecasting models into the Flask app.

May 27: Finish AI analysis panel integration using OpenRouter/OpenAI APIs.

May 29: Complete full MVP integration and core feature testing.

May 31: Finish bug fixes, UI polishing, deployment setup, and final presentation preparation.

June 1: Final project submission and deployment.

Completion Criteria

- All MVP features (auth, dashboard, stock viewer, forecast, portfolio/watchlist, AI analysis, relevant equations) are implemented and reachable by navbar
- Frontend, Flask backend, and SQLite database are fully integrated
- ML forecast page returns predictions for any of the FAANG tickers with a confidence interval
- AI Analysis returns a LLM-generated response to at least 3 test prompts
- App served on financebros.app

Open Questions

- Do we want OpenRouter to call a specific LLM by default or allow users to choose between different models?
- Should ML forecasts predict only next-day prices or support multi-day forecasting?
- Do we want cached stock data to refresh automatically on a timer or only when requested?
- Should the watchlist and portfolio system support simulated profit/loss tracking?
- Do we want to support candlestick charts if the MVP is completed early?
- Should AI analysis responses include links/news sources or only generated summaries?
- Do we want guest access without login for basic stock searching and charts?
- How much historical stock data should be stored locally in SQLite before clearing old cache entries?
- Should the application prioritize desktop dashboard layouts or mobile responsiveness first?
- Do we want to deploy locally only or fully host the project online for public access?

Appendix

- Bloomberg Terminal pricing is estimated at approximately \$25,000 per year, which creates a large accessibility barrier for students and beginner investors.
- Live and oldyFinance data to improve ML forecasting accuracy.
- OpenRouter was selected because it supports multiple LLM providers under one API and includes student developer credits.
- SQLite was chosen over MongoDB/PostgreSQL because the project mainly reads cached data and does not require large-scale concurrent writes.
- Chart.js was selected because it is lightweight, easy to integrate with Flask, and supports line graphs, comparison charts, and volume visualizations needed for the MVP.
- The project intentionally excludes real-money trading, cryptocurrency trading, options trading, and high-frequency live market streaming to keep the scope manageable.
- Stretch goals such as paper trading, heatmaps, and candlestick charts will only be implemented if all MVP functionality is stable before the deadline.
- The project is intended primarily for educational and learning purposes rather than professional financial advising or investment management.

Other

N/A